

Bruno Wasserstein

RenovAI: a generative artificial intelligence platform for residential renovation visualisation - a computational solution and business plan for the Brazilian market

SÃO PAULO
2026

Bruno Wasserstein

RenovAI: a generative artificial intelligence platform for residential renovation visualisation - a computational solution and business plan for the Brazilian market

Final Course Project submitted to the
Institute of Technology and Leadership
(INTELI) as a partial requirement to obtain
a Bachelor's degree in Computer Science.

Advisor: Prof. Murilo Zanini de Carvalho

Cataloging in Publication
Library and Documentation Service
Institute of Technology and Leadership (INTELLI)
Data entered by the author.

Sobrenome, Nome

Título do trabalho: subtítulo / Nome Sobrenome do autor; Nome e
Sobrenome do orientador. – São Paulo, 2025.

nº de páginas : il.

Trabalho de Conclusão de Curso (Graduação) – Curso de [Ciência da
Computação] [Engenharia de Software] [Engenharia de Hardware] [Sistema
de Informação] / Instituto de Tecnologia e Liderança.

Bibliografia

1. [Assunto A]. 2. [Assunto B]. 3. [Assunto C].

CDD. 23. ed.

ACKNOWLEDGEMENTS

First and foremost, I would like to thank my advisor, Prof. Murilo Zanini de Carvalho, for his guidance, trust, and encouragement throughout this project. Even when the idea still seemed like a cloud, unclear, turbulent, and without a defined direction, he believed in my potential and helped me find alternative paths. He encouraged me not to fear the time spent studying, researching, and preparing before execution, reminding me that, many times, we must first build a strong foundation before attempting to raise a skyscraper. Beyond his role as an advisor, he became an example of passion for technology, intellectual rigour, and dedication to research. In every discussion, he showed me that arguments should prevail over emotion, and that good ideas must be built, questioned, and defended with clarity. Above all, he was an example of what it means to be a teacher, a mentor, and a true master.

I would also like to thank my older brother, Gustavo, who is my greatest example in life and one of my greatest professional role models. He battles every day for what he dreams of and spares no effort to reach the places he wants to reach. Throughout this project, he advised me, challenged my assumptions, brought new insights, and created productive conflicts of ideas that pushed me to take RenovAI to another level. More than that, he has always been an example of dedication, excellence, and responsibility, consistently delivering beyond what was expected and doing everything with remarkable care, precision, and commitment.

I am deeply grateful to my grandmother, Amalia. Since I was a child, she has supported me in my studies and in life, always encouraging me to learn, research, question, and pursue knowledge. She taught me to value science, curiosity, and education not only as academic tools, but as ways of understanding and transforming the world. She is also one of my greatest academic examples: at a time when many believed that women should not pursue higher education, she challenged the expectations of her generation and graduated from one of the most prestigious law schools in Brazil. Without her support, example, and encouragement, I would not have come to know Inteli, nor would I have gone as far as I once believed possible.

I would also like to thank my aunt, Silvia, for always supporting me and encouraging me to seek success while remaining generous, empathetic, and committed to helping others. She showed me that empathy is a strength, and that

understanding another person's pain is often the first step toward creating something meaningful. In many ways, this lesson made RenovÁl possible.

I thank my father for supporting me throughout this journey and for teaching me to keep researching, studying, and seeking knowledge. He helped me understand the value of work, persistence, and responsibility. Through his work, he also awakened my interest in architecture and engineering, encouraging me to learn more, explore these fields, and see how ideas can become real structures, spaces, and solutions. In many ways, his example helped inspire the path that eventually led me to RenovÁl.

I thank my mother, without whom I might never have returned to university. She has always cared deeply for my wellbeing and showed me the value of helping others. Through her example, I learned that when we teach, support, and serve others, we also learn and grow ourselves. In many ways, RenovÁl is also about this exchange: helping users make better decisions while learning from their needs, doubts, and experiences.

I would also like to thank my master and life mentor, Alexandre Luke. With his wisdom, he taught me never to stop fighting, and helped me reflect on mission, values, goals, and the importance of thinking about others. His guidance has been a constant reminder that ambition must be connected to purpose.

I thank Paulo Marcos, my older brother figure, friend, and mentor. I remember being fourteen years old and listening to him tell me stories about his work at Cabify. Over time, those stories became lessons from different professional journeys, challenges, and adventures he lived through. Through them, he showed me that dedication is one of our greatest tools, that we must always be willing to learn, and that we do not need to know everything in order to move forward. What matters is understanding what we need to know, continuing to learn and evolve, and having the humility to ask for help from those who know more than we do. He taught me that the arrogance of believing we do not need help is one of our greatest flaws. For the past ten years, he has helped me think about business, life, goals, and work, always encouraging me to seek improvement, question my ideas, pursue knowledge, and become better without ever harming others in the process.

I would also like to thank my uncle, Henrique, who was a second father to me and the first mentor I ever had in life. I still remember being around ten or eleven years

old and hearing him tell me that I could be very intelligent, but that I needed to learn how to build focus, because only through focus would I be able to achieve my goals. His words, affection, and concern stayed with me throughout the years. He was the first person to encourage me to seek the discipline necessary to reach my full potential. He taught me not to doubt myself, but also not to become arrogant, reminding me that true growth requires humility, constant learning, and the willingness to keep improving.

I thank the Institute of Technology and Leadership (Inteli) for the education, environment, and opportunities that made this work possible. I also thank the professors who supported me throughout this journey, from the most ambitious and unconventional ideas to the more tangible projects. Each of them, in different ways, helped me understand the value of research, discipline, and intellectual rigour.

I thank the early users of RenovAI, whose feedback, doubts, and reactions helped shape the product and made this project more grounded in real needs.

Finally, I thank my friends, both those who have been with me for years and those whom life brought closer to me in recent months. Some believed in this idea from its earliest and most uncertain moments, while others arrived in the final stretch and helped me not only keep going, but also aim higher, deliver more, and believe more deeply in what I was building. Their support, encouragement, ideas, and trust in me became a source of strength during one of the most demanding phases of this journey. More than simply helping me continue, they brought me happiness, and that happiness became fuel to keep fighting with even more determination, courage, and purpose.

“Recognising the need is the primary condition for design.” (Charles Eames)

RESUMO

WASSERSTEIN, Bruno. RenovAI: uma plataforma de inteligência artificial generativa para a visualização de reformas residenciais. 2026. 19 f. Trabalho de Conclusão de Curso (Graduação) - Ciência da Computação, Instituto de Tecnologia e Liderança (INTELI), São Paulo, 2026.

A RenovAI é uma plataforma digital brasileira de inteligência artificial generativa para o planejamento de reformas residenciais, voltada ao público geral e não a arquitetos profissionais. A partir de uma descrição ou fotografia de um cômodo, a plataforma gera uma imagem fotorrealista do ambiente reformado e uma lista de móveis sugeridos, em poucos minutos e sem custo. O trabalho apresenta o desenvolvimento da solução computacional, baseada em tecnologias web modernas e em serviços de inteligência artificial generativa em nuvem, e o respectivo plano de negócios. O problema endereçado é a incerteza antes do gasto: 82% das moradias brasileiras são feitas sem profissional habilitado (CAU/Datafolha, 2022) e o projeto de arquitetura custa de R\$ 55 a R\$ 140 por metro quadrado. O modelo de negócios baseia-se em comissões sobre vendas, com planos pagos opcionais. A validação técnica foi conduzida com usuários reais em produção. Conclui-se que a solução é tecnicamente robusta e que o modelo é promissor, condicionado à validação das hipóteses de conversão com dados próprios.

Palavras-chave: inteligência artificial generativa; design de interiores; reforma residencial; plano de negócios; plataforma SaaS.

“Recognising the need is the primary condition for design.” (Charles Eames)

ABSTRACT

WASSERSTEIN, Bruno. RenovAI: a generative artificial intelligence platform for residential renovation visualisation. 2026. 19 p. Final Course Project (Bachelor) - Computer Science, Institute of Technology and Leadership (INTELI), São Paulo, 2026.

RenovAI is a Brazilian generative artificial intelligence platform for residential renovation planning, aimed at the general public rather than professional architects. From a description or photograph of a room, the platform generates a photorealistic image of the renovated space and a list of suggested furniture, in minutes and at no cost. This work presents the development of the computational solution, based on modern web technologies and cloud generative-AI services, and the corresponding business plan. The problem addressed is uncertainty before spending: 82% of Brazilian homes are built without a qualified professional (CAU/Datafolha, 2022), and an architecture project costs from R\$ 55 to R\$ 140 per square metre. The business model is based on commissions on sales, with optional paid plans. Technical validation was carried out with real users in production. The solution is found to be technically robust and the model promising, contingent on validating the conversion hypotheses with proprietary data.

Keywords: generative artificial intelligence; interior design; residential renovation; business plan; SaaS platform.

LIST OF ILLUSTRATIONS

Figure 1 - RenovAI result - original room (left) and AI-generated renovation (right)

LIST OF TABLES

Table 1 - Project overview

Table 2 - Market sizing (TAM, SAM, SOM)

Table 3 - Customer personas

Table 4 - Competitive comparison (overview)

Table 5 - SWOT matrix

Table 6 - SWOT matrix (continued)

Table 7 - Business Model Canvas

Table 8 - Key performance indicators monitored

LIST OF ABBREVIATIONS AND ACRONYMS

AI Artificial Intelligence

MVP Minimum Viable Product

TAM/SAM/SOM Total / Serviceable Available / Serviceable Obtainable Market

BMC Business Model Canvas

CDN Content Delivery Network

LGPD Brazilian General Data Protection Law

SWOT Strengths, Weaknesses, Opportunities and Threats

SUMMARY

1	Introduction.....	11
2	Solution Development.....	12
2.1	Definition of Market Assumptions and Hypotheses	12
2.2	Market Sizing and Analysis.....	13
2.3	Competitive Analysis and Differentials	14
2.4	Technological Solution	15
2.5	The Business Plan.....	16
2.6	Validation and Results.....	17
3	Conclusion.....	18
	References	19
	Appendices.....	19
	Annexes	19

1 Introduction

Deciding to renovate or furnish a home is, for most Brazilians, a high-stakes financial decision made without technical vocabulary and without the budget to hire an architect. RenovAI is a generative artificial intelligence platform that lets anyone visualise a renovated room before spending a single cent, addressing a structural gap in access to interior-design services in Brazil.

Context and Motivation.

Architecture and interior-design services in Brazil remain elitist. A residential architecture project costs on average between R\$ 55 per square metre (GetNinjas, 2024) and R\$ 140 per square metre in renovations (Habitissimo, 2024) - equivalent to thousands of reais for a single room. As a result, only 18% of Brazilians hire an architect or engineer for their own construction work, meaning 82% of homes are built without a qualified professional (CAU/Datafolha, 2022). Generative AI created, for the first time, the technical possibility of democratising this access in a scalable way, at a marginal cost of a few cents per visualisation.

Problem Definition and Value Proposition.

The core problem is not aesthetics but uncertainty before spending: the fear of getting a renovation wrong. RenovAI's value proposition is to deliver visual certainty before the purchase decision - a photorealistic image of the renovated space plus a curated, priced furniture list - for free, on a mobile phone, in minutes, requiring no technical knowledge from the user.

Objectives of the Work.

The general objective is to develop and validate a computational solution for AI-assisted residential renovation and to elaborate a business plan for its introduction into the market. The specific objectives are:

- build a functional MVP that generates a photorealistic visualisation and a furniture list from a user briefing;
- validate the product technically with real users in production;
- define a revenue model for the solution;
- elaborate a complete business plan, including market sizing, competitive analysis and financial feasibility.

Justification and Contributions.

The addressable market - annual household spending on home goods in Brazil - is estimated at R\$ 92 billion. Beyond the economic relevance, the solution has documented social-impact potential aligned with the United Nations Sustainable Development Goals 1, 10 and 11, by lowering the cost of design decisions for low-income households that renovate the home they already have. The technical contribution is a production-grade, fully localised generative pipeline for the Brazilian market (prices in reais, Brazilian brands, native Portuguese).

Structure of the Work.

Chapter 1 introduces the problem and objectives. Chapter 2 develops the solution: it defines the market assumptions and hypotheses (2.1), sizes and analyses the market (2.2), examines the competition (2.3), describes the technological solution (2.4), presents the business plan (2.5) and reports the market validation and results (2.6). Chapter 3 concludes the work. The References, Appendices and Annexes close the document.

Table 1 - Project overview

Aspect	Description
Product stage	MVP in production at renovai.ia.br
Target audience	General public planning a residential renovation
Business model	Commission on sales, with optional paid plans
Reference market	Home-goods spending by Brazilian urban households
Operating structure	100% digital, cloud-native, lean fixed cost
Funding	Bootstrapped by the founder; non-dilutive public grants under consideration

Source: prepared by the author (2026).

2 Solution Development

This chapter develops the RenovAI solution end to end. It begins with the strategic assumptions and hypotheses that guided the product (2.1), quantifies the market opportunity (2.2) and the competitive landscape (2.3), details the technological solution actually built and running in production (2.4), consolidates the business plan (2.5), and reports the validation carried out with real users (2.6).

2.1 Definition of Market Assumptions and Hypotheses

The product was designed around three explicit, falsifiable hypotheses, each to be confirmed or refuted with proprietary data collected from real usage.

Problem Hypothesis.

Most people planning a renovation face a real pain point - the fear of buying wrong - and do not hire a professional, given the cost. The assumption is that this public will adopt a free tool that delivers visual certainty before they spend.

Solution Hypothesis.

The assumption is that a pure generative-AI solution - instant, mobile-first, with a furniture list linked to real stores - is a better way to solve the problem than the existing alternatives (human-in-the-loop services that cost from R\$ 149 and take days, or web-only AI tools without a shoppable list).

Value Hypothesis.

The assumption is that the revenue model is acceptable to users: the visualisation is free, monetised indirectly through commissions on furniture the user would buy anyway, with optional paid plans for heavier users.

2.2 Market Sizing and Analysis

Market size was estimated bottom-up, with each factor traceable to a published primary source.

Market Size (TAM, SAM, SOM).

The Total Addressable Market is annual spending by Brazilian urban households on home goods. The Serviceable Available Market restricts it to households with purchasing power and service consideration. The Serviceable Obtainable Market is RenovAI's conservative three-year potential, an internal assumption to be validated.

Table 2 - Market sizing (TAM, SAM, SOM)

Layer	Value	Methodology / source
TAM	R\$ 92 billion/year	67.7 million urban households (FJP, 2023) × R\$ 1,362 average home-goods spend (ABCasa/IEMI, 2024)
SAM	R\$ 30.8 billion/year	TAM × 45.7% (classes A-C1, ABEP, 2024) × 73% (would consider an architect, CAU/Datafolha, 2022)
SOM (year 3)	R\$ 8.5 million/year	0.5% penetration of the SAM × hybrid ARPU of R\$ 75/year - internal assumption to validate

Source: prepared by the author (2026), based on the cited sources.

Customer Segmentation and Personas.

RenovAI serves several customer segments, from mass-market renovators to higher-income owners of compact properties, in addition to local artisans as future marketplace suppliers.

Table 3 - Customer personas

Persona	Profile	Job to be done
A - Decided Renovator	Adult, middle class, renovating a room	Certainty before spending
B - Urban Dweller	Young adult, first apartment	Translate inspiration into a decision
C - Compact High-End Owner	Young high-income, compact flat	Quality result fast, no waiting
D - Local Artisan	Artisan, future marketplace supplier	Reach buyers with a defined project

Source: prepared by the author (2026).

2.3 Competitive Analysis and Differentials

A mapping conducted in May 2026 identified 27 relevant players. The Brazilian AI-assisted interior-design market is more crowded than a first read suggests, but the segments differ.

Direct and Indirect Competitors.

Direct Brazilian competitors include both AI-only tools and human-assisted services (such as Arquiteto de Bolso, DecoraGPT, IAdecora and Archie). International references that frame the category include Wayfair (with its Decorify/Muse AI tools, but absent from Brazil), Houzz, RoomGPT and Havenly, while the closure of the pure-human Mody illustrates that human-in-the-loop models struggle to scale.

Table 4 - Competitive comparison (overview)

Criterion	RenovAI	AI-only tools	Human-assisted services
Price	Free (MVP)	Free / paid	From R\$ 149
Delivery	AI, under a minute	AI, minutes	Human, days
Localisation (PT-BR, BRL)	Yes	Varies	Yes
Mobile-first experience	Yes	Mostly web	Mostly web

Source: prepared by the author (2026).

Competitive Advantage and Differentiating Factor.

Generative AI itself is an increasingly commoditised resource; RenovAI differentiates through deep localisation for the Brazilian market (prices in reais, local brands, native Portuguese), a mobile-first experience and ease of use for non-experts, supported by a low operating cost.

2.4 Technological Solution

RenovAI is a 100% digital, serverless platform in production at renovai.ia.br.

Requirements and Specifications.

The main functional requirements are: user authentication; a five-step briefing wizard; AI generation of a photorealistic image and a furniture list; project history; budget export to PDF. The main non-functional requirements are: generation in under 60 seconds; high availability through model redundancy; cost control via a per-user generation quota; data protection compliant with the Brazilian LGPD (encrypted personal identifiers, row-level security).

Architecture and Technology.

The frontend is built with modern web technologies (React, TypeScript) and hosted on a global CDN. The backend uses serverless functions and a managed PostgreSQL database with row-level security on every table. Image generation relies on a leading commercial generative-AI service, with automatic fallback to alternative providers to ensure availability even if a single provider fails. Figure 1 shows a representative result.

Figure 1 - RenovAI result - original room (left) and AI-generated renovation (right)



Source: RenovAI platform (renovai.ia.br), 2026.

Development and Implementation (MVP).

Development followed an agile, sprint-based methodology with daily deployment, supported by AI-assisted engineering. The MVP, in production since April 2026, implements authentication, project creation and management, AI image generation, furniture identification and recommendation, project storage and history, and a draft Retailer Portal for business-to-business partners.

Testing and Technical Evaluation.

Robustness is ensured by provider redundancy in the generation pipeline, rate limiting and usage quotas for cost control. The platform has generated 67 projects in production with completed generation, validating the end-to-end pipeline (briefing interpretation, image generation, furniture extraction and storage).

2.5 The Business Plan

The business plan consolidates the market, business model, go-to-market and financial feasibility of RenovAI at a strategic level.

Market and Competitor Analysis (SWOT).

Table 5 - SWOT matrix

Strengths	Weaknesses
Native localisation for the Brazilian market (language, prices, brands); mobile-first experience; low operating cost; product already in production.	Early-stage product; revenue model still under validation; lean team; dependence on third-party AI services.

Source: prepared by the author (2026).

Table 6 - SWOT matrix (continued)

Opportunities	Threats
Public housing-renovation credit programmes; fast-growing AI design segment; falling cost of generative AI; a large category validated internationally yet under-served in Brazil; non-dilutive public funding.	Entry of better-funded competitors; commoditisation of generative AI; exchange-rate pressure on cloud costs; restrictions in some retail affiliate programmes.

Source: prepared by the author (2026).

Business Model (Business Model Canvas).

Table 7 - Business Model Canvas

Block	Content
Customer Segments	General-public renovators across income levels; retailers and developers (B2B).
Value Propositions	See your renovation before buying: photorealistic visualisation + furniture list, free, in minutes, no technical knowledge.
Channels	Mobile-first web (renovai.ia.br); organic SEO and social media; referral; retail partnerships.
Customer Relationships	Self-service product; conversational assistant; e-mail support; native sharing.
Revenue Streams	Commissions on referred sales; optional paid plans; future marketplace.
Key Resources	Generative-AI pipeline; product and engineering; user base with saved projects.

Key Activities	Product development; AI integration; partnership onboarding; curation.
Key Partnerships	Cloud and AI providers; furniture retailers; potential public funding agencies.
Cost Structure	Variable cloud-AI usage and a lean fixed-cost base; no inventory or logistics.

Source: prepared by the author (2026).

Marketing and Sales Strategy.

The go-to-market is fully digital, with no human sales force. Acquisition combines organic channels (search, social media and referral) with retail and developer partnerships. Paid media is introduced only once the commission revenue can sustain the cost of acquisition.

Financial Projection and Feasibility.

The model combines a low, cloud-native fixed-cost base with commission-based revenue and optional subscriptions. The variable cost per use is modest and has been falling with the broader decline in generative-AI costs. Feasibility depends on validating user conversion with proprietary data; the lean cost base provides runway to do so. The initial investment was bootstrapped by the founder, with non-dilutive public grants under consideration.

2.6 Validation and Results

Validation followed a lean, evidence-based approach, using the product in production with a controlled user base to test the business hypotheses rather than to claim market traction.

Validation Methodology.

The MVP was released in production and instrumented with funnel analytics (sign-up, wizard completion, successful generation, sharing, PDF export) under explicit LGPD consent, with UTM attribution by channel.

Market Validation Results.

As of May 2026 the platform had 20 registered users and 8 users who generated at least one project, totalling 67 completed generations. The activation rate (sign-up to first generation) was 40%, identifying the activation funnel as the main bottleneck. These figures constitute deliberate technical validation, not statistical market traction.

Pivoting or Persisting.

Validation refined the revenue assumptions and led to the introduction of optional paid plans alongside the commission model, while preserving the core thesis rather than abandoning it.

Table 8 - Key performance indicators monitored

KPI	What it measures	How to measure
Registered users	Reach	Database
Active users	Real product usage	Database
Activation rate	Sign-up to first generation	Funnel analytics
Click-through on products	Purchase intent	Link tracking
Conversion to paid plan	Monetisation	Subscription data

Source: prepared by the author (2026).

Risks and Mitigation Plan.

- **Financial:** the revenue model requires validation with proprietary data; mitigated by a lean cost base and optional paid plans.
- **Technological:** dependence on third-party AI services; mitigated by provider redundancy in the pipeline.
- **Competitive:** a well-funded incumbent could enter; mitigated by localisation and product focus.
- **Legal:** data protection (LGPD) and tax treatment of revenue; mitigated by basic compliance already implemented and planned legal review.

3 Conclusion

This work developed and validated RenovAI, a generative-AI platform that lets any Brazilian visualise a renovated room and obtain a furniture list for free, and elaborated the corresponding business plan. The objectives were met: a functional MVP is in production, validated technically with 67 generated projects; a revenue model was defined; and a complete business plan, including market sizing, competitive analysis and financial feasibility, was produced.

The central finding is that the solution is technically robust and the business model is feasible, contingent on validating user conversion with proprietary data. The platform's lean, cloud-native cost base provides runway for that validation.

Future work includes formalising the company, validating the revenue assumptions with real data, expanding the product across platforms, and growing the user base.

References

ASSOCIAÇÃO BRASILEIRA DE EMPRESAS DE PESQUISA (ABEP). Critério Brasil 2024. São Paulo: ABEP, 2024. Available at: <https://abep.org/>. Accessed on: 15 jun. 2026.

ASSOCIAÇÃO BRASILEIRA DE ARTIGOS PARA CASA, MESA E PRESENTES (ABCASA); IEMI. Setor de artigos para casa movimenta R\$ 102 bilhões em 2024. São Paulo: ABCasa/IEMI, 2024. Available at: <https://www.ecommercebrasil.com.br/noticias/setor-de-artigos-para-casa-movimentar-102-bi-em-2024>. Accessed on: 15 jun. 2026.

CONSELHO DE ARQUITETURA E URBANISMO DO BRASIL (CAU/BR); DATAFOLHA. Percepções sobre Arquitetura e Urbanismo. Brasília: CAU/BR, 2022. Available at: <https://www.cau.br.gov.br/pesquisa2022/>. Accessed on: 15 jun. 2026.

FUNDAÇÃO JOÃO PINHEIRO (FJP). Déficit habitacional no Brasil. Belo Horizonte: FJP, 2024. Available at: <https://fjp.mg.gov.br/deficit-habitacional-no-brasil/>. Accessed on: 15 jun. 2026.

GETNINJAS. Preço de projeto residencial. 2024. Available at: <https://www.getninjas.com.br/>. Accessed on: 15 jun. 2026.

HABITISSIMO. Orçamentos de arquitetos. 2024. Available at: <https://www.habitissimo.com.br/>. Accessed on: 15 jun. 2026.

WAYFAIR INC. Wayfair launches Decorify, a virtual room styler powered by generative AI. Investor Relations, 2023. Available at: <https://investor.wayfair.com/>. Accessed on: 15 jun. 2026.

Appendices

No appendices are included in this public version of the work.

Annexes

No annexes (third-party documents) are included in this version.